

Math 350: Homework 2

Each problem is worth 10 points. Please show all your work and explain your answers fully for full credit.

1. For each of the following, give three examples of this property, and then prove it is true.
 - (a) If n is an even integer, then $-n$ is an even integer.
 - (b) If n is an odd integer, then $-n$ is an odd integer.
 - (c) If n is an even integer, then $(-1)^n = 1$. You may use standard properties of exponents.
2. Prove the second identity in DeMorgan's Law; that is, suppose A and B are subsets in $\mathbb{R}$. Show
$$(A \cap B)^c = A^c \cup B^c.$$
3. For sets A , B and C and universal set U , draw the Venn diagram representing the following:
 - (a) $(A \setminus B)^c$
 - (b) $A \cup (B \setminus C)$
 - (c) $(A \cap B) \setminus C$
 - (d) $A^c \cap (B \setminus C)$
4. Suppose A, B and C are sets with $C \neq \emptyset$. Prove that if $A \times C = B \times C$, then $A = B$. Why is it necessary that we specify that $C \neq \emptyset$?